GROUP B9 · FINAL REVIEW · STUDY GUIDE FOR PERSON C

Hybrid A* and NMPC: From the General Idea to Our Truck

Part 3 of the presentation: how the planner finds a route the truck can drive, how the predictive controller follows it without breaking the hitch limit, and, in full detail, what the NMPC cost function means term by term.

Your slides: 14–17 Your time: about 5 minutes From Person B (kinematics) · to Person D (results)

1. The whole project in one page
2. What your part must achieve
3. Planning and control: why two layers
4. Graph search and A*, from scratch
5. Why grid A* fails for a truck
6. Hybrid A*: the idea
7. Our motion primitives and the hitch regulator
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1 The Whole Project in One Page

Read this first. Every presenter should be able to tell this story in one minute, because the professor may ask any of you how your part fits in.

The Problem

Cars such as the Mahindra BE 6 and XEV 9e can already park themselves (Autopark: 12 ultrasonic sensors, perpendicular, angled and parallel bays). A truck with a trailer cannot, and it is the vehicle that needs help most. The reason is physics: a tractor-trailer is two bodies joined at a hitch, and **when it reverses, the hitch angle grows on its own**. A 2° error becomes 60° after 28 m of reversing. If nobody corrects it, the trailer folds against the cab: a **jackknife**.

What We Built

A guidance system that takes a tractor-trailer from the yard gate into a chosen bay while the hitch angle never exceeds 60°. It has four layers:

Layer	What it does	Presented in
Vehicle model	Four numbers describe the rig; four equations say how they change for a given speed and steering	Part 1 (A), Part 2 (B)
Hybrid A* planner	Searches the yard once and returns a route the truck can actually drive, gate to bay	Part 3 (C)
NMPC controller	Every 0.25 s, predicts 9.6 s ahead and chooses the steering that follows the route with $\|\psi\| \le 60°$ as a hard limit	Part 3 (C)
Driver guidance	Turns the controller's output into gear, speed (km/h), steering-wheel turns and distance	Part 4 (D)

The Result

We tested 110 free bays (4 layouts × 6 random fills of parked trucks). All 110 were parked within tolerance, the hitch limit was never broken (worst case 50.1°), and a driver model without hitch awareness jackknifed on all 72 reversing manoeuvres.

The Chain of Ideas

THE STORY IN FOUR SENTENCES

(1) We model the truck as a wheeled robot with four state variables and two inputs. (2) Its kinematics show that reversing is unstable: the hitch angle grows exponentially. (3) So we plan a drivable route with Hybrid A* and follow it with an NMPC that looks ahead and treats the hitch limit as a hard constraint. (4) The result parks every test bay and tells a human driver exactly what to do.

Symbol	Meaning	Value
x_1, y_1	Position of the tractor's rear-axle centre	metres
$\theta_0,\ \theta_1$	Heading of the tractor, heading of the trailer (from the x axis, anticlockwise positive)	radians or degrees
$\psi = \theta_0 - \theta_1$	Hitch angle: how far the tractor is turned relative to the trailer	hard limit 60°
$v,\ \delta$	Speed (negative in reverse), front-wheel steering angle	$\|v\| \le 2$ m/s, $\|\delta\| \le 35^\circ$
$L_1,\ L_2,\ d$	Tractor wheelbase, hitch to trailer axle, hitch ahead of the rear axle	4 m, 8 m, 0.5 m

2 What Your Part Must Achieve

Person B ends with: "reversing is unstable, so a controller must look ahead". You answer: *here is how we decide where to go (Hybrid A*), and here is how we steer to get there safely (NMPC)*. For each algorithm, the professor asked for two things: what it is in general, and how it ties to our project. Leave him with three ideas:

1. **Hybrid A*** is A* search whose moves are short, drivable arcs simulated on the real vehicle model, so every path it returns can actually be driven.

2. **NMPC** predicts the next 9.6 s with the same model, chooses the steering that best follows the plan, applies only the first bit, and repeats every 0.25 s.
3. The NMPC's **cost** says what "following well" means (mainly: keep the trailer axle on the plan and the hitch near its planned angle, steer smoothly), and its **hard constraints** say what is never allowed ($|\psi| > 60^\circ$, hitting anything).

3 Planning and Control: Why Two Layers

	Planner (Hybrid A*)	Controller (NMPC)
Question	Which way should the truck go?	How should it steer right now?
Looks at	The whole yard: walls, parked trucks, the bay	The next 9.6 s of the plan
Runs	Once per manoeuvre (and to re-plan the final approach)	Every 0.25 s
Strength	Finds a way around obstacles, decides where to stop and switch gear	Corrects drift and disturbances, enforces the hitch limit
Alone it fails because	It is open loop: in reverse any small error grows (Person B)	It is local: it cannot find a way around a parked truck

4 Graph Search and A*, From Scratch

Path planning as a search: cut the world into **nodes** (for a grid, each free cell), connect neighbours with **edges** that have a cost (length), and find the cheapest chain of edges from start S to goal G.

4.1 The Algorithm

A* keeps an **open list** of nodes waiting to be expanded, ordered by

$$f(n) = g(n) + h(n)$$

- $g(n)$: the cost of the best path found so far from S to n (known exactly).
- $h(n)$: the **heuristic**, an estimate of the cost still to go from n to G.
- $f(n)$: estimated total cost of the best path through n.

1. Put S on the open list with $g = 0$.
2. Pop the node with the **smallest** f. If it is G, stop and trace the path back through parent links.
3. Otherwise mark it **closed** and, for each neighbour, compute $g_{\text{new}} = g(n) + \text{edge cost}$. If that beats the neighbour's best g so far, record n as its parent and push it with $f = g_{\text{new}} + h$.
4. Repeat from step 2.

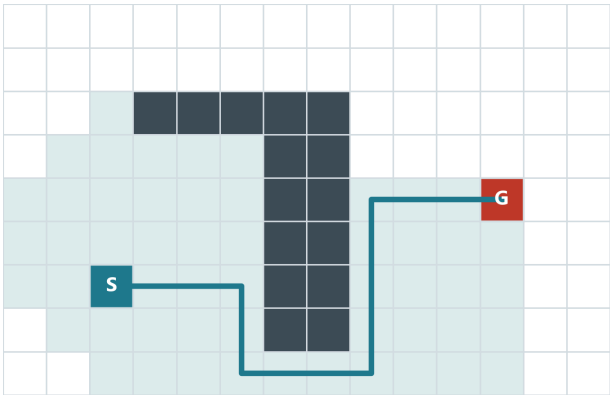


Figure 1. A* on a 14 × 9 grid (4-connected, Manhattan heuristic). Shaded: the 55 cells it expanded. Line: the shortest path, 15 steps. Without h (Dijkstra) it would expand cells in every direction.

4.2 The Heuristic

If h **never overestimates** the true remaining cost (it is *admissible*), A* returns the cheapest path. If $h = 0$ it becomes Dijkstra's algorithm: correct but slow. A good heuristic is as close to the true cost as possible without exceeding it.

Weighted A* uses $f = g + \varepsilon h$ with $\varepsilon > 1$. It is greedier, expands far fewer nodes, and its path costs at most ε times the optimum. We use $\varepsilon = 3$: speed matters more to us than the last few metres of path length.

5 Why Grid A* Fails for a Truck

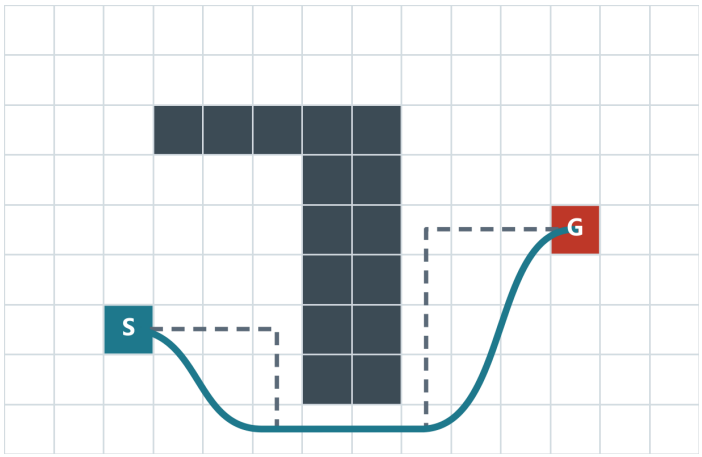


Figure 2. Dashed: grid A* path with sharp corners. Solid: a smooth path a vehicle can follow.

- A grid path has 90° corners; a 12 m rig cannot turn on the spot (minimum rear-axle radius 5.7 m).
- Grid cells ignore heading and hitch angle, but where the truck can go next depends on both (nonholonomic, Person A).
- It ignores reversing, gear changes and the 60° hitch limit.

6 Hybrid A*: the Idea

Hybrid A* (Dolgov, Thrun et al., Stanford, for the DARPA Urban Challenge) keeps A*'s open list and $f = g + h$, but changes what a node and a move are:

	Grid A*	Hybrid A*
Node stores	A cell	An **exact continuous state**; here $(x_1, y_1, \theta_0, \psi)$
Moves	Jump to a neighbouring cell	**Simulate a short drivable arc** (a motion primitive) on the vehicle model
Grid used for	Everything	Only to spot duplicates: two states in the same cell count as the same node
Result	Undrivable corners	Every path is drivable, because it was driven in simulation

Our cell size (for duplicate detection only): 0.5 m in position, 5° in θ_0, 12° in ψ, plus the gear. Hybrid A* is *resolution complete* in practice: it finds a path if one exists at this resolution, but it is not guaranteed optimal, because snapping to cells discards some states.

The Analytic Shot

Arcs alone rarely land exactly on the goal pose. So, from nodes reasonably close to the bay, the planner also tries a direct **shot**: a Dubins path (shortest curve with bounded curvature) from the trailer axle to the bay, driven on the real model by a hitch controller. The shot is accepted only if it is collision-free and ends inside the bay (within 0.35 m laterally, 4° heading, 5° hitch). Often the search is: a few arcs to get lined up, then one shot backwards into the bay.

7 Our Motion Primitives and the Hitch Regulator

A motion primitive for a car is "drive 2 m with the wheel at angle δ". For our truck that fails in reverse: with a fixed wheel the hitch angle runs away (Person B), so a reverse arc would end in a jackknife after a few expansions. Instead, each primitive **steers the hitch to a target angle** ψ_{des} with a small feedback law.

7.1 Deriving the Regulator (Feedback Linearisation)

We *choose* how we want the hitch to behave: decay towards the target at a rate k per metre, in either gear:

$$\dot{\psi} = -|v|\, k\, (\psi - \psi_{\text{des}})$$

Person B's factored hitch equation is $\dot{\psi} = v\left[\frac{\tan\delta}{L_1}\left(1 - \frac{d}{L_2}\cos\psi\right) - \frac{\sin\psi}{L_2}\right]$. Set the two equal and solve for $\tan\delta$ (note $|v| = v\,\text{sign}(v)$):

$$\tan\delta = L_1 \frac{\dfrac{\sin\psi}{L_2} - \text{sign}(v)\, k\, (\psi - \psi_{\text{des}})}{1 - \dfrac{d}{L_2}\cos\psi}$$

Check by substituting back: the bracket becomes $\frac{\sin\psi}{L_2} - \text{sign}(v)k(\psi - \psi_{\text{des}}) - \frac{\sin\psi}{L_2}$, the unstable $\sin\psi$ term cancels exactly, and $\dot{\psi} = -v\,\text{sign}(v)\,k(\psi - \psi_{\text{des}}) = -|v|k(\psi - \psi_{\text{des}})$. ✓

With $k = 0.5$ per metre the error shrinks ×e every 2 m, forward *or* reverse. The steering is clipped to 85 % of the lock (29.75°) so the controller keeps some margin.

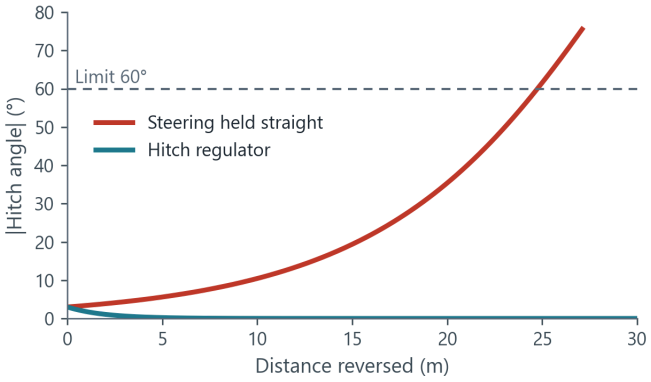


Figure 3. Reversing from a 3° hitch angle (our model): with the wheel straight it passes 60° after about 25 m; with the regulator it returns to 0 within a few metres.

7.2 The Ten Primitives

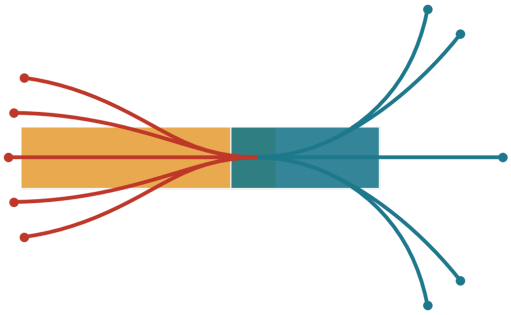


Figure 4. From one pose: five forward arcs (teal) and five reverse arcs (red).

- 2 gears × 5 hitch targets $\psi_{\text{des}} \in \{0^\circ, \pm 25^\circ, \pm 45^\circ\}$ = **10 arcs per node**.
- Each arc is 2 m long, integrated with RK4 in 4 steps of 0.5 m.
- An arc is **discarded** if the hitch exceeds 52° (a margin below the 60° limit) or any point of either body comes within 0.35 m of an obstacle.

8 Our Planner's Cost and Heuristic

8.1 Cost of One Arc (Adds to g)

$$c = \underbrace{2\,\text{m} \times \begin{cases} 1.0 & \text{forward} \\ 1.4 & \text{reverse} \end{cases}}_{\text{distance}} + \underbrace{0.25\,|\psi_{\text{des}}| \times 2}_{\text{sharp turns}} + \underbrace{0.6\,|\psi_{\text{des}} - \psi_{\text{des}}^{\text{prev}}|}_{\text{changing turns}} + \underbrace{0.6\,\frac{|\psi_{\text{end}}|}{52^\circ} \times 2}_{\text{bent hitch}} + \underbrace{6}_{\text{if gear changes}}$$

(angles in radians). Read it as preferences: shorter is better; reversing costs 40 % more than driving forward; gentle, steady turns beat sharp or changing ones; a straight hitch beats a bent one; every gear change costs as much as 6 m of driving. The planner then finds the route with the lowest total.

8.2 The Heuristic

$$h = 3 \times \Big(\max\big(d_{\text{Dubins}},\ d_{\text{grid}}\big) + 3\,|\psi| \Big)$$

- d_{Dubins}: length of the shortest Dubins path of the **trailer axle** to its parked pose, with turning radius 10 m. Why the trailer axle? It is a *flat output*: its path curvature κ alone fixes the hitch angle, $\psi = \arctan(L_2\kappa)$.

So the trailer axle moves like a car whose "steering" is ψ, and a 10 m radius corresponds to $\psi = \arctan(8/10) = 38.7^\circ$.

- d_{grid}: shortest distance to the bay around obstacles on a 0.5 m grid (ignores heading). The max of the two is a better estimate than either.
- $3|\psi|$: a bent hitch still has to be straightened.
- The factor 3 is the weighted-A* ε: plans in about 0.1 to 6 s instead of minutes.

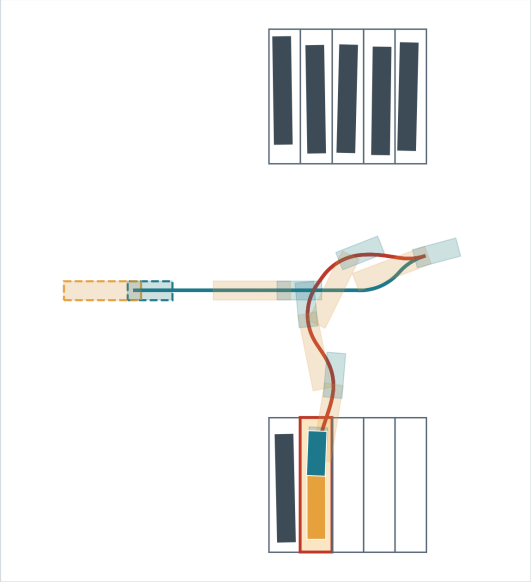


Figure 5. A real plan: drive in forward (teal), stop, reverse into the bay (red), with rig footprints along the way.

9 Model Predictive Control, From Scratch

MPC is control by repeated planning over a short horizon. At every control step:

1. **Measure** the current state X_0.
2. **Predict**: for any candidate sequence of future inputs, simulate the model N steps ahead.
3. **Optimise**: find the input sequence that minimises a **cost** J while satisfying **constraints**.
4. **Apply only the first input**, wait one control period, and go back to step 1. The horizon slides forward each time: a **receding horizon**.

"Nonlinear" MPC means step 2 uses the full nonlinear model (Person B's equations), not a linear approximation. The general problem is

$$\min_{u_1,\ldots,u_M} \; J(X,u) \quad \text{subject to} \quad X_{k+1} = f(X_k,u_k), \quad g(X_k,u_k) \le 0$$

SAY IT LIKE THIS

"MPC is like a chess player: it thinks several moves ahead, picks the best line, plays only the first move, then thinks again after seeing the reply. Because it looks ahead, it sees a jackknife coming while there is still time to steer away."

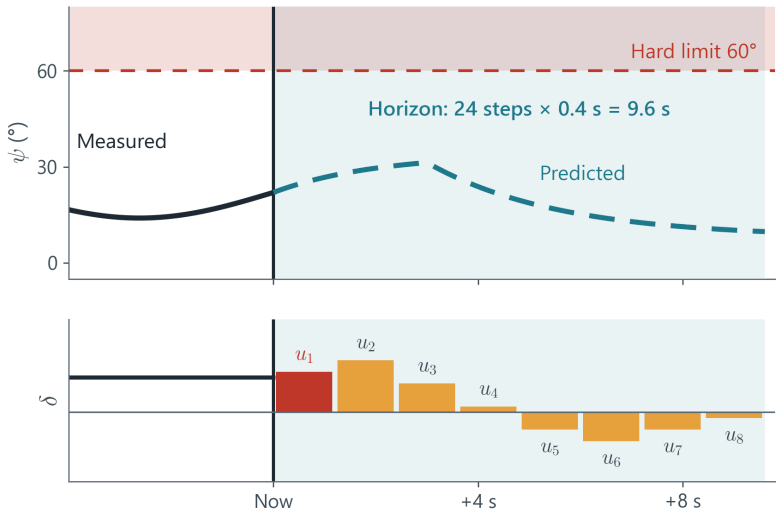


Figure 6. Our horizon: measured hitch angle up to now, predicted over the next 9.6 s (24 steps of 0.4 s), below the 60° limit. Bottom: the 8 steering moves $u_1 \dots u_8$; only u_1 is applied, for 0.25 s.

10 Our NMPC Problem

Item	Our choice
Prediction	24 steps × 0.4 s = **9.6 s**, RK4 on the four-state model
Decision variables	**8 steering corrections** $u_1 \dots u_8$, each held for 3 steps (1.2 s)
Steering actually applied	$\delta_k = \mathrm{clip}\left(\delta^{\mathrm{ff}}_k + u_{\lceil k/3 \rceil},\ \pm 35^\circ\right)$, where δ^{ff} is the planned (feed-forward) steering
Speed	From the plan's speed profile; **not** a decision variable (the guidance tells the driver how fast)
Reference X^{ref}_k	The planned state at the distance the rig is predicted to reach at step k
Control period	Re-solved every **0.25 s**

Why corrections around the plan? The plan was driven on the same model, so its steering δ^{ff} is already nearly right. $u = 0$ means "follow the plan exactly"; the optimiser only has to find small corrections for drift and disturbances. **Why 8 blocks instead of 24 values?** Fewer variables solve much faster, and a driver could not follow a steering wheel that changes every 0.4 s anyway.

11 The Cost Function, Term by Term

This is the part to understand best. The slide shows the compact form:

$$J = \sum_{i=1}^{24} \left\| X_i - X_i^{\text{ref}} \right\|_Q^2 \; + \; \sum_{j=1}^{8} \left(R\, u_j^2 + S\, (u_j - u_{j-1})^2 \right)$$

It has three ideas: **tracking** (stay on the plan), **effort** (do not correct more than needed) and **smoothness** (do not jerk the wheel). The optimiser looks for the 8 corrections that make J as small as possible.

11.1 What $\|e\|_Q^2$ Means

For an error vector e and a diagonal weight matrix $Q = \text{diag}(q_1, q_2, \dots)$:

$$\|e\|_Q^2 = e^T Q\, e = q_1 e_1^2 + q_2 e_2^2 + \cdots$$

It is a weighted sum of squared errors. Each weight q_i says how much we care about that error compared with the others. Example: $e = (0.3, 0.05)$ and $Q = \text{diag}(0.6, 15)$ give $0.6 \times 0.09 + 15 \times 0.0025 = 0.054 + 0.0375 = 0.0915$.

11.2 Which Errors We Track

Error at prediction step i	Units	Meaning
e_{p_2}: trailer-axle position minus planned	m	Where the back of the rig is compared with the plan (a 2-D vector, both x and y)
e_{θ_1}: trailer heading minus planned	rad	Is the trailer pointing the planned way?
e_{p_1}: tractor rear-axle position minus planned	m	Where the cab is compared with the plan
e_{θ_0}: tractor heading minus planned	rad	Is the cab pointing the planned way?
e_{ψ}: hitch angle minus planned	rad	Is the rig bent as much as planned, no more?

Angle errors are wrapped into $(-180^\circ, 180^\circ]$ so that 359° and 1° count as 2° apart.

11.3 The Full Formula With Our Real Weights

Written out exactly as the code computes it (all angles in radians):

$$
\begin{aligned}
J \;=\; & \sum_{i=1}^{24} \Big(0.6\,\|e_{p_2,i}\|^2 + 1.5\,e_{\theta_1,i}^2 + 0.15\,\|e_{p_1,i}\|^2 + 0.3\,e_{\theta_0,i}^2 + 15\,e_{\psi,i}^2 \Big) && \text{stage cost} \\
& + \Big(8\,\|e_{p_2,24}\|^2 + 60\,e_{\theta_1,24}^2 + 1.0\,\|e_{p_1,24}\|^2 + 60\,e_{\psi,24}^2 \Big) && \text{terminal cost} \\
& + 0.02 \sum_{j=1}^{8} u_j^2 && \text{effort } (R) \\
& + 3 \sum_{j=1}^{8} (u_j - u_{j-1})^2 && \text{smoothness } (S)
\end{aligned}
$$

Here u_0 is the correction that was applied in the previous solve, so the first change is also measured against what the wheel is doing now.

11.4 Why Each Weight Is What It Is

Term and weight	Reason
Trailer axle position 0.6, heading 1.5	The **main target**. The trailer axle is the flat output and the part that must end up in the bay. Tracking the tractor's axle instead fails in reverse, because the tractor first moves the wrong way (non-minimum phase, Person B Section 10).
Hitch angle 15	The **largest stage weight**. Staying close to the planned hitch angle keeps the rig away from the unstable region; when forced to choose, we prefer a straighter rig to a slightly tighter path.
Tractor position 0.15, heading 0.3	Small: keep the cab sensible, but let it deviate if that helps the trailer.
Terminal: 8, 60, 1.0, 60	The horizon must **end on the plan**, aligned and with the planned hitch. Without it the optimiser could leave a growing error just beyond the horizon, which in reverse is exactly where trouble starts.
Effort R = 0.02	Tiny: corrections are cheap, because the plan's steering may need real adjustment. It only breaks ties in favour of "follow the plan".
Smoothness S = 3	Large compared with R: change the wheel **gradually**, so a human driver can follow and the steering (limited to 45°/s) can keep up.

11.5 Why Squares?

- **Sign does not matter**: 0.3 m left and 0.3 m right are equally bad.
- **Big errors cost much more** than small ones: twice the error, four times the cost. The controller fixes large deviations first and tolerates tiny ones.
- **Smooth**: a quadratic has a gradient everywhere, which the solver (SQP, Section 13) needs, and near the optimum the problem looks like a bowl, so it converges in a few iterations.

11.6 Worked Example: One Prediction Step

Suppose at one step the trailer axle is 0.3 m off the plan, the trailer and tractor headings are 2° off, the tractor axle is 0.3 m off, and the hitch is 3° off. In radians, $2^\circ = 0.0349$ and $3^\circ = 0.0524$.

Term	Calculation	Cost
Trailer position	0.6×0.3^2	0.054
Trailer heading	1.5×0.0349^2	0.0018
Tractor position	0.15×0.3^2	0.0135
Tractor heading	0.3×0.0349^2	0.0004
Hitch	15×0.0524^2	0.041
Stage cost for this step		**0.111**

Reading it: 0.3 m of trailer error and 3° of hitch error cost about the same (0.054 and 0.041). That is the balance the weights were tuned for: metres and radians are different units, and the weights make them comparable. Heading errors of 2° are cheap on their own; they matter because they turn into position errors a few steps later, which the cost then sees.

Effort and smoothness in the same units: a 5° correction (0.0873 rad) costs $0.02 \times 0.0873^2 = 0.00015$: almost free. Changing the correction by 5° from one block to the next costs $3 \times 0.0873^2 = 0.023$, about the same as a 2.2° hitch error held for one step. So the controller corrects readily, but spreads a large correction over several blocks.

11.7 What Happens If You Change a Weight

If you increase	The controller will
Hitch weight (15)	Keep the rig closer to its planned bend; may track the path a little more loosely
Trailer position weight (0.6)	Hug the path harder; may bend the hitch more to do it
Smoothness S (3)	Turn the wheel more gently; reacts more slowly to disturbances
Effort R (0.02)	Stay closer to the planned steering; corrects less
Terminal weights	Insist that the horizon ends on the plan; steadier, but can be over-cautious

Only the **ratios** between weights matter: multiplying every weight by 10 gives the same optimal steering.

12 Hard Constraints Versus Cost

Some things must never happen, whatever the cost says. These are **hard constraints**, checked at every one of the 24 predicted steps:

$$|\psi_i| \le 60^\circ \qquad \text{clearance}_i \ge 0.25\ \text{m} \qquad |u_j| \le 35^\circ,\ \ \delta_i \text{ clipped to } \pm 35^\circ$$

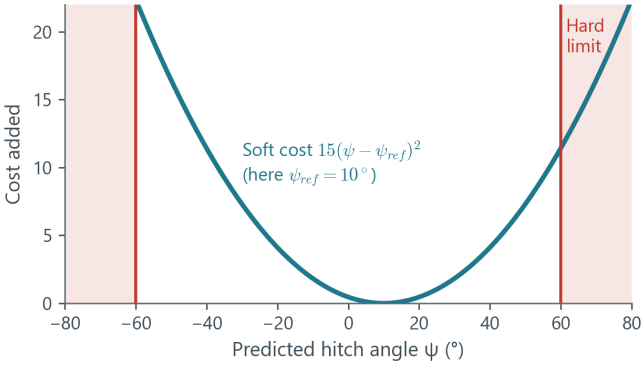


Figure 7. The hitch term of the cost (a soft preference, a bowl around the planned angle) against the hard limit (a wall at ±60° the solver may not cross).

Why not just a big penalty? A penalty can always be outweighed: if tracking the path badly enough costs more than the penalty, the optimiser would accept a 65° hitch. A constraint cannot be traded away. In our tests the hitch never exceeded 50.1°, against the 60° limit. (Our benchmark also ran the NMPC with the hitch limit as a soft cost only, to show the difference.)

In the code the hitch limit is written as 48 inequalities ($60^\circ - \psi_i \geq 0$ and $60^\circ + \psi_i \geq 0$ for each step), and clearance as 24 inequalities using a signed-distance map of the yard over both bodies.

13 How the Problem Is Solved

- **Single shooting:** only the 8 corrections are variables. For any guess, the predicted states come from simulating the model, so the dynamics are satisfied automatically.
- **SLSQP** (Sequential Least-Squares Quadratic Programming, from SciPy): each iteration approximates the cost by a quadratic and the constraints by straight lines, solves that small problem, and moves towards its answer. It repeats until it stops improving (at most 40 iterations).
- **Gradients** by finite differences: nudge each of the 8 corrections slightly and re-simulate. We simulate all 9 versions in one batched computation, which is why a solve takes 19 to 42 ms, far inside the 250 ms control period.
- **Warm start:** each solve starts from the previous solution, which is usually almost right.
- **Fallbacks:** if no solution satisfies everything, try again keeping only the hitch limit hard (a rig that cannot both dodge and stay unfolded must stay unfolded). If that fails too, the stable hitch regulator from Section 7 steers for that step.
- **Horizon stops at the segment end:** the prediction holds still where the planned segment ends (at a gear change or in the bay), so a wall behind the bay is not mistaken for a collision the rig would never reach.

14 How the Two Work Together

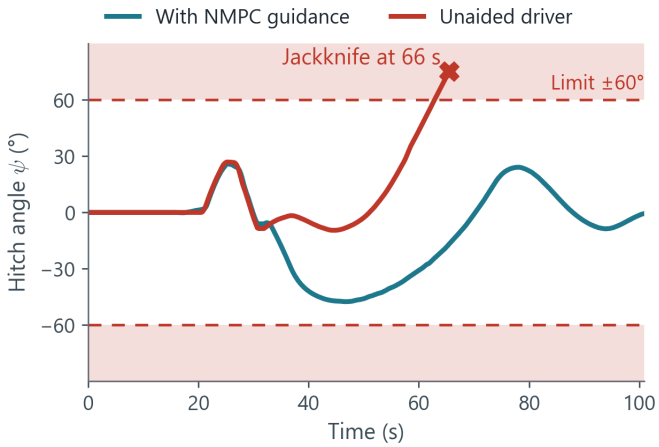


Figure 8. One real bay from our tests. With NMPC (teal) the hitch swings to about −48° on purpose to turn the trailer, then comes back. The unaided driver (red), on the same planned path, passes 60° and jackknifes at 66 s.

1. The user picks a free bay. **Hybrid A*** plans the route once: forward set-up, stop, reverse into the bay.
2. The route is cut at gear changes into segments, each with a planned steering δ^{ff} and speed profile.
3. Every 0.25 s the **NMPC** solves for the steering correction, with the hitch limit hard.
4. The **guidance layer** turns the steering into wheel turns, km/h and distance for the driver.
5. Near the bay the final approach is re-planned from the actual state every 8 m, so small errors never accumulate.

15 What to Say on Each Slide

Slide 14 · Part 3 divider (10 s)
"Person B showed that reversing is unstable. Now: how we plan a route the truck can drive, and how we follow it without jackknifing."

Slide 15 · Hybrid A*: Planning a Path the Truck Can Drive (about 1.5 min)
Left: A* with $f = g + h$; grid paths have corners. Hybrid A*: exact poses, expanded by drivable arcs. Right: our state includes ψ; 10 arcs of 2 m, each steering the hitch to a target; drop arcs over 52° or within 0.35 m; heuristic is the trailer axle's Dubins path.

Slide 16 · NMPC: Predict, Optimise, Apply, Repeat (about 2 min)
The three steps. Our horizon 9.6 s, 8 moves. Then the cost at the bottom: tracking, effort, smoothness; trailer axle and hitch weighted most. Hard limits: 60°, 35°, 0.25 m. SLSQP, 19 to 42 ms, every 0.25 s.

Slide 17 · How Hybrid A* and NMPC Work Together (about 1 min)
Pipeline left to right. Planner global and once; controller local and every 0.25 s; why each alone fails. Hand over: "Person D will show the results and the limitations."

16 Questions the Professor May Ask

What is the difference between A* and Dijkstra?

Dijkstra orders by g only; A* by $g + h$. A good heuristic focuses the search towards the goal. With $h = 0$ they are the same.

Is your heuristic admissible? Is the path optimal?

No and no, deliberately. We multiply it by 3 (weighted A*) to plan in seconds instead of minutes, which gives up the optimality guarantee; in practice the routes are close to the shortest. What we do guarantee is that every route is drivable and collision-free, because it was simulated and checked.

Why Hybrid A* and not RRT?

Hybrid A* is deterministic (same yard, same plan), handles a cost (prefer forward, few gear changes) naturally, and works well in structured spaces like parking yards. RRT is random and usually needs smoothing.

Why do the primitives use a hitch regulator?

With a fixed steering angle, reverse arcs are unstable and jackknife within a couple of expansions. The regulator cancels the unstable term (feedback linearisation) so the hitch converges to its target in both gears.

Why is the trailer axle used in the heuristic and the cost?

It is a flat output: its curvature fixes ψ, so it behaves like a car steered by ψ. And in reverse the tractor's axle is non-minimum phase; tracking it makes the controller steer the wrong way first.

Why NMPC rather than PID or LQR?

It handles hard constraints, uses the full nonlinear model, and looks ahead, which the unstable reverse dynamics need. PID only reacts; LQR has no constraints and is linear.

How did you choose the 9.6 s horizon?

The hitch error grows ×e every 8 m in reverse; at reversing speed that takes several seconds. The horizon must be long enough to see the growth coming while there is still time to act; a much shorter horizon sees the jackknife too late.

What do Q, R, S do?

Q weights tracking errors (trailer axle and hitch most), R penalises correction size (small), S penalises changes of correction (large, for smooth steering). Only their ratios matter.

Why a terminal cost?

So the predicted motion must end on the plan; otherwise the optimiser could push an error just past the horizon, where in reverse it would grow.

Is the NMPC guaranteed stable?

We have no formal proof (no terminal set). We certify empirically: every bay is driven in closed loop and checked before it is offered, and pushes of ±10° on the hitch are caught by the live controller.

What if SLSQP fails?

Retry with only the hitch limit hard; if still infeasible, the hitch regulator steers. In the 110-bay tests the hitch limit was never broken.

Why only 8 decision variables?

Fewer variables solve far faster (we need under 250 ms), and steering held 1.2 s at a time is something a human driver can actually follow.

17 Check Yourself

1. A node has $g = 12$ and $h = 5$; another has $g = 9$ and $h = 9$. Which does A* expand first? Which does weighted A* with $\varepsilon = 3$ expand first?
2. What is the cost of one reverse arc with $\psi_{\text{des}} = 25^\circ$ that follows an arc with $\psi_{\text{des}} = 0$, ends with $\psi = 20^\circ$ and does not change gear?
3. With the regulator ($k = 0.5$ per m), how far must the rig drive for a 10° hitch error to fall to about 1.35°?
4. Compute $15\, e_\psi^2$ for a 6° hitch error. How does it compare with a 0.5 m trailer-axle error?
5. Why would multiplying Q, R and S all by 10 not change the steering?
6. In one sentence each: what is the difference between a soft cost and a hard constraint?

Answers

1. A*: $f = 17$ vs $f = 18$, so the first. Weighted: $12 + 15 = 27$ vs $9 + 27 = 36$, still the first.
2. Distance $2 \times 1.4 = 2.8$; sharp turn $0.25 \times 0.436 \times 2 = 0.218$; change $0.6 \times 0.436 = 0.262$; bent hitch $0.6 \times (20/52) \times 2 = 0.462$; no gear change. Total ≈ 3.74.
3. The error falls ×e every 2 m; $10^\circ \to 1.35^\circ$ is a factor $7.4 \approx e^2$, so about 4 m.
4. $6^\circ = 0.1047$ rad, $15 \times 0.01097 = 0.164$. A 0.5 m trailer error costs $0.6 \times 0.25 = 0.15$: about the same.
5. The minimiser of J and of $10J$ is the same set of corrections; only the ratios between terms decide the trade-off.
6. A soft cost makes something expensive but allows it if the rest of the cost gains more; a hard constraint forbids it whatever the cost.

18 One-Page Cheat Sheet

A* and Hybrid A*

$f = g + h$; pop the lowest f. Admissible $h \Rightarrow$ optimal; we use $\varepsilon = 3$ for speed. Hybrid A*: exact states $(x_1, y_1, \theta_0, \psi)$, moves are 10 simulated 2 m arcs (2 gears × $\psi_{\text{des}} \in \{0, \pm 25^\circ, \pm 45^\circ\}$), cells 0.5 m × 5° × 12° only for duplicates. Drop arcs with $|\psi| > 52^\circ$ or clearance < 0.35 m. Analytic Dubins shot into the bay.

Hitch Regulator

$\tan\delta = L_1 \dfrac{\sin\psi / L_2 - \text{sign}(v) k (\psi - \psi_{\text{des}})}{1 - (d/L_2)\cos\psi} \Rightarrow \dot{\psi} = -|v| k (\psi - \psi_{\text{des}})$, k = 0.5 /m: ×e every 2 m, both gears.

Planner Cost and Heuristic

Arc: $2 \times (1.0 \text{ fwd} \mid 1.4 \text{ rev}) + 0.25|\psi_{\text{des}}| \cdot 2 + 0.6|\Delta\psi_{\text{des}}| + 0.6\frac{|\psi|}{52^\circ} \cdot 2 + 6$ per gear change. $h = 3(\max(d_{\text{Dubins}}, d_{\text{grid}}) + 3|\psi|)$, trailer-axle Dubins radius 10 m.

NMPC

24 × 0.4 s = 9.6 s horizon; 8 corrections held 1.2 s; $\delta = \delta^{\text{ff}} + u$; re-solve every 0.25 s; SLSQP, 19–42 ms; warm start; fallbacks: hitch-only, then regulator.

Cost

Stage: $0.6\|e_{p_2}\|^2 + 1.5e_{\theta_1}^2 + 0.15\|e_{p_1}\|^2 + 0.3e_{\theta_0}^2 + 15e_{\psi}^2$. Terminal: $8, 60, 1, 60$. Effort $R = 0.02$, smoothness $S = 3$. Trailer axle and hitch matter most; smooth wheel; end on the plan.

Hard Constraints

$|\psi| \le 60^\circ$, clearance ≥ 0.25 m, $|\delta| \le 35^\circ$ at every predicted step. Cannot be traded against tracking. Worst case in tests: 50.1°.